Appendix

5.1 Counterfactuals

Counterfactual reasoning is a concept from psychology and social sciences [23], which involves examining possible alternatives to past events [31]. Humans often use counterfactual reasoning by imagining what would happen if an event had not occurred, and this is precisely what counterfactual reasoning is. When applied to artificial intelligence, the question is, for example, "Why did the model make this decision instead of another?" (counterfactual explanation) or "How would the decision have differed if a certain condition had been changed?". This reasoning can take the form of a counterfactual or semi-factual explanation.

A counterfactual explanation might be "If your income had been $10000 higher, then your credit would have been accepted" [19].A semi-factual is a special-case of the counterfactual in that it conveys possibilities that "counteract" what actually happened, even if the outcome does not change [2]: "Even if your income had been $5000 higher, your credit would still be denied" (but closer to being accepted). Within the framework of counterfactual reasoning, we here used the method described in [20] and the notion of trajectory. Indeed, the study of counterfactual trajectories a posteriori is of great interest, as it also makes it possible to identify when a trajectory is approaching the frontier or pass trough the frontier. In this paper, the trajectory of a counterfactual is the minimum of changes in the input vector to change the predicted class to the opposite[1]. In the case of Sepsis, one could be interested, for example, when examining Patients :

- predicted as negative (Target=0) but close to the frontier; what are their counterfactual (Target=1) ? The reasons that will cause them to be paid in the opposite case?
- predicted as positive (Target=1) but for which one could be interested in finding reasons to transform them predicted as negative?

We give below one example of these two possibilities when one considers all the input variables (of the input vector) as "alterable". But as explained in the literature and in [20] it is up to the user to limit the list of the alterable variables since some of them could not be changed. This must be done in consultation with the expert of the application domain, here a patrician.

In the first case :

- for the patient '10001' (Patient_Id) : Initial class = '0' - Proba '0' = 0.561962
- the trajectory for the counterfactual as a single stage :
  - Step 1 : if "StdDev(EtCO2)" change of value from 5.795418449 to 0.2537

After the first stage, the initial predicted probability (0.561962) to be negative becomes 0.131336. Therefore, he becomes positive.

In the second case

- for the patient '100003' (Patient_Id) : Initial class = '1' - Probability '1' = 0.999787

[1] A notebook that performs this process for the Khiops library is given in the Github for reproducibility

- the trajectory for the counterfactual as 6 stages :
  - Step 1 : 'Min(Temp)' from 37.6 to 31.3135 → Proba '1' = 0.999021,
  - Step 2 : 'Max(HospAdmTime)' from -253.56 to 8.022499 → Proba '1' = 0.994351,
  - Step 3 : 'Min(pH)' from 7.398 to 7.269 → Proba '1' = 0.972920,
  - Step 4 : 'StdDev(EtCO2)' from 2.965952629 to '8.17991062' → Proba '1' = 0.912302,
  - Step 5 : 'Median(FiO2)' from 0.4 to 1.0095 → Proba '1' = 0.751472,
  - Step 6 : 'Median(HospAdmTime)' from 253.56 to 8.022499 → Proba '1' = 0.471857,

  After the 6th stage, the initial predicted probability (0.561962) to be positive becomes 0.471857. Therefore, the patient becomes negative.

The information provided by the trajectory for the counterfactual is clearly beneficial to understand and to act on the positivity of the Sepsis.

5.2 Detailed results versus the used classifier

Legend :

- Rob Auc: Robustness on the AUC (AUC_{Test} / AUC_{Train}))
- Size: #Var used by "RF or CB or XGB" for a the same value of Q / #Var used by khiops for a given value of Q
- CatBoost did not succeed in dealing with the 100000 Variables on the computer used (HP EliteBook 655 15.6 inch G10 Notebook PC, AMD Ryzen 5 PRO 7530U with Radeon Graphics, 2000 MHz, 6 cores(s), 12 processeur(s) logique(s), RAM 32Go)
- LSTM: One hidden layer, H: Number of neurons, Optimiser: Adam, Learning Rate: 0.001, Number of epochs=100

Khiops								
Q	R	S=#Var	Acc Train	Auc train	Acc Test	Auc Test	Rob AUC	Size
10	5	5	0.7835	0.8647	0.7716	0.8444	0.98	1
100	86	44	0.9332	0.9813	0.9132	0.9737	0.99	1
1000	872	98	0.9530	0.9913	0.9360	0.9833	0.99	1
10000	8849	198	0.9703	0.9962	0.9337	0.9819	0.99	1
100000	72331	272	0.9916	0.9991	0.9406	0.9784	0.98	1
Random Forest								
Q	R	S=#Var	Acc Train	Auc train	Acc Test	Auc Test	Rob AUC	Size
10	-	9	1.0	1.0	0.7831	0.8281	0.83	1.80
100	-	99	1.0	1.0	0.9155	0.9757	0.98	2.25
1000	-	996	1.0	1.0	0.9383	0.9817	0.98	10.16
10000	-	8964	0.9819	0.9985	0.9360	0.9785	0.98	45.27
100000	-	38743	0.9357	0.9841	0.9086	0.9725	0.99	142,43
CatBoost								
Q	R	S=#Var	Acc Train	Auc train	Acc Test	Auc Test	Rob AUC	Size
10	-	9	0.8906	0.9575	0.7899	0.8592	0.90	1.80
100	-	99	0.9964	0.9999	0.9269	0.9802	0.98	2.25
1000	-	860	0.9992	0.9999	0.9406	0.9850	0.98	8.77
10000	-	2662	1.0	1.0	0.9406	0.9833	0.98	13.44
XgBoost								
Q	R	S=#Var	Acc Train	Auc train	Acc Test	Auc Test	Rob AUC	Size
10	-	9	0.9934	0.9997	0.7899	0.8527	0.85	1.80
100	-	88	1.0	1.0	0.9246	0.9786	0.98	2.0
1000	-	607	1.0	1.0	0.9452	0.9847	0.98	6.19
10000	-	1159	1.0	1.0	0.9360	0.9834	0.98	5.85
100000	-	1176	1.0	1.0	0.9337	0.9832	0.98	4.32
LSTSM								
-	-	34	0.908	0.967	0.877	0.936	0.97	H=128
-	-	34	0.920	0.972	0.874	0.945	0.97	H=64
-	-	34	0.913	0.967	0.877	0.932	0.96	H=32
-	-	34	0.894	0.953	0.852	0.920	0.97	H=16

Table 5: Results versus the number of Q variables to be constructed and classifiers (Train 90%, Test 10%.)